Yiming Xu

Department of Civil Engineering and Engineering Mechanics
Columbia University
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ACADEMIC APPOINTMENTS

Columbia University **Nov 2026 - Present**
Postdoctoral Research Scientist, Department of Civil Engineering and Engineering Mechanics *New York, NY*
Advisor: Andrew Smyth

The University of Texas at Austin **Sep 2023 - Oct 2026**
Postdoctoral Fellow, School of Architecture & Dept. of Civil, Architectural & Environmental Engineering *Austin, TX*
Advisors: Junfeng Jiao and Christian Claudel

University of Florida **Aug 2019 - Aug 2023**
Research Assistant, Department of Civil and Coastal Engineering *Gainesville, FL*

EDUCATION

University of Florida **Aug 2019 - May 2023**
Ph.D. in Civil Engineering *Gainesville, FL*
Advisor: Xilei Zhao
Committee members: Lily Elefteriadou, Siva Srinivasan, Daisy Wang
Dissertation: AI-enabled Travel Demand Forecasting for Shared Mobility

Tongji University **Sep 2016 - Jun 2019**
M.S. in Transportation Engineering *Shanghai, China*
Advisor: Jian Sun
Thesis: An Importance Sampling Approach for High-Risk Scenario Reconstruction and Accelerated Testing of Autonomous Vehicles

Tongji University **Sep 2012 - Jun 2016**
B.S. in Transportation Engineering *Shanghai, China*
Advisor: Jian Sun

RESEARCH INTERESTS

My research integrates **artificial intelligence** and **transportation systems** to advance safe, efficient, equitable, and sustainable urban mobility. My interests include:

- **AI for transportation**: Spatiotemporal learning, explainable AI, and language and vision-language models.
- **Travel behavior and demand modeling**: Mode choice, travel patterns, activity-based models, and demand forecasting.
- **Digital twins and AI agents**: Multilayer digital twins, generative AI agents, and agent-based simulation.
- **Emerging and sustainable mobility**: Shared mobility, micromobility, electric vehicles, and aviation electrification.
- **Transportation safety and resilience**: Crash analysis, safety evaluation, and evacuation modeling.
- **Robotics and physical AI**: Autonomous vehicles, delivery robots, embodied intelligence, and world models.

RESEARCH EXPERIENCE

Postdoctoral Research Scientist **Oct 2026 - Present**
Department of Civil Engineering and Engineering Mechanics
Columbia University

- Smart Streetscapes: Research Thrust 2 - Situational Awareness, 2026 - Present, funded by the NSF Center for Smart Streetscapes (CS3)

Postdoctoral Fellow **Sep 2023 - Oct 2026**
School of Architecture & Dept. of Civil, Architectural & Environmental Engineering
The University of Texas at Austin

- Large Language Models for Human-Centered Mobility Planning, 2026
- Neuro-Symbolic Edge AI for Autonomous Vehicle Safety: A Knowledge-Graph Guided VLM, 2026
- LLM-Driven Travel Behavior Modeling: Retrieval-Augmented Generation for Predicting Trip Dynamics, 2025 - 2026
- Extended Reality-Enhanced Digital Twin for Agentic Mobile Robots, 2025 - 2026
- Emergency Medical Service Arrival Time Estimation and Route Optimization using Traffic Digital Twin, 2025 - 2026
- Digital Infrastructure and Transportation Digital Twins for Monitoring, Operations, and Safety, 2025 - 2026
- Agent-SUMO: Integrating SUMO with AI Agents for Interactive Urban Mobility Simulation, 2025 - 2026
- Data-Driven Modeling of E-Scooter Demand and Multimodal Interactions with Ridesourcing, 2023 - 2025, funded by USDOT Tier 1 UTC CM2 and UT Good Systems
- Advancing Electric Vehicle Safety, Equity, and Infrastructure Planning for Sustainable Urban Mobility, 2024 - 2025, funded by NSF and UT Good Systems
- Multilingual AI-Assisted Emergency Preparedness: Enhancing Resilience and Equity in Underserved Communities, 2024 - 2025, funded by the City of Austin
- Autonomous Vehicle Safety in the Smart City Era, 2024, funded by UT Good Systems
- Digital Twin as Catalyst for Sustainable and Smart City, 2023 - 2024, funded by USDOT Tier 1 UTC CCST
- Austin Digital Twin: Integrated 3D Modeling and Multi-Layered Data Analysis, 2023 - 2024, funded by UT Good Systems
- Navigating LLM Ethics, 2023 - 2024, funded by NSF and UT Good Systems

Research Assistant **Aug 2019 - Aug 2023**
Department of Civil and Coastal Engineering
University of Florida

- Shared Micromobility Demand Forecasting with Deep Learning, 2021 - 2023, funded by USDOT Regional UTC: STRIDE and the UF AI Research Catalyst Fund
- Evacuation Behavior Analysis using Large-Scale GPS Data, 2020 - 2022, funded by NIST
- Micromobility as a Solution to Reduce Urban Traffic Congestion, 2019 - 2022, funded by USDOT Regional UTC: STRIDE
- Interpretable Machine Learning on the Adoption of Ride-splitting, 2019 - 2021, funded by USDOT Regional UTC: STRIDE

Research Assistant **Jun 2018 - Aug 2018**
Department of Civil and Environmental Engineering
University of Massachusetts Lowell

- Vehicle Cooperation Around Lane-Changing, 2018

Research Assistant **Sep 2016 - Jun 2019**
School of Transportation Engineering
Tongji University

- Data-Driven Accelerated Testing Scenario Construction for Autonomous Vehicle Safety Evaluation, 2017 - 2019
- Vehicle Turning Behavior Modeling at Mixed-Flow Intersections, 2016 - 2018

INDUSTRY EXPERIENCE

Research Intern **Sep 2022 - Dec 2022**

DiDi Labs, Mountain View, CA

- Developed a motion prediction model incorporating GCN, VAE, and attention operations.
- Evaluated the proposed model using the Waymo Open Dataset. The proposed model achieved a 13% improvement in prediction accuracy and a 70% improvement in trajectory diversity compared with the baseline Multipath++ model.

PUBLICATIONS

Peer-Reviewed Journal Papers

(** Corresponding author.* † *Co-first authors; equal contribution.*)

First-Author and Corresponding-Author Publications

1. **Xu, Y.***, Jiao, J. (2026). Benchmarking Retrieval-Augmented Generation Strategies for Large Language Model-Based Travel Mode Choice Prediction. *Travel Behaviour and Society*.
2. **Xu, Y.***, Ke, Q., Zhang, X., Zhao, X. (2025). ICN: Interactive Convolutional Network for Forecasting Travel Demand of Shared Micromobility. *GeoInformatica*.
3. Chio, S., **Xu, Y.***, Jiao, J. (2025). Utility or Equity? A Critical Analysis of Existing Public Electric Vehicle Charger Allocations in Austin, Texas. *Transportation Research Part D: Transport and Environment*.
4. Xu, N., **Xu, Y.***, Liu, J., Jiao, J. (2025). How Do EV Crashes Differ from ICEV Crashes: A Comparative Study of Pennsylvania. *Journal of Safety Research*.
5. Jiao, J., **Xu, Y.***, Li, Y. (2024). Exploring Spatial Heterogeneity of E-scooter's Relationship with Ridesourcing using Explainable Machine Learning. *Transportation Research Part D: Transport and Environment*.
6. Jiao, J., **Xu, Y.*** (2024). Analyzing Shared E-Scooter Trip Frequency on Urban Road Segments in Austin, TX. *Case Studies on Transport Policy*.
7. **Xu, Y.**, Zhao, X.*, Zhang, X., Paliwal, M. (2023). Real-Time Forecasting of Dockless Scooter-Sharing Demand: A Spatio-Temporal Multi-Graph Transformer Approach. *IEEE Transactions on Intelligent Transportation Systems*.
8. **Xu, Y.**, Zhao, X.*, Lovreglio, R., Kuligowski, E., Nilsson, D., Cova, T. J., Yan, X. (2022). A Highway Vehicle Routing Dataset During the 2019 Kincade Fire Evacuation. *Scientific Data*.
9. **Xu, Y.**, Yan, X., Sisiopiku, V., Merlin, L., Xing, F., Zhao, X.* (2022). Micromobility Trip Origin and Destination Inference Using General Bikeshare Feed Specification Data. *Transportation Research Record*.
10. **Xu, Y.**, Yan, X., Liu, X., Zhao, X.* (2021). Identifying Key Factors Associated with Ridesplitting Adoption Rate and Modeling Their Nonlinear Relationships. *Transportation Research Part A: Policy and Practice*.
11. **Xu, Y.**, Ma, Z., Sun, J.* (2019). Simulation of Turning Vehicles' Behaviors at Mixed-Flow Intersections Based on Potential Field Theory. *Transportmetrica B: Transport Dynamics*.
12. **Xu, Y.**, Zou, Y., Sun, J.* (2018). Accelerated Testing for Automated Vehicles Safety Evaluation in Cut-In Scenarios Based on Importance Sampling, Genetic Algorithm and Simulation Applications. *Journal of Intelligent and Connected Vehicles*.

Co-Authored Publications

1. Wang, T., Zeng, T., Zeng, Z., Zhang, F., Wang, Y., Li, X., **Xu, Y.**, Chen, S., Jiao, J., Claudel, C., Chen, X.* (2026). DBPnet: Damper Characteristics-Based Bayesian Physics-Informed Neural Network for Wheel Load Estimation. *IEEE Transactions on Vehicular Technology*.
2. Wang, H., Davis, W., **Xu, Y.**, Yu, J., Mai, G., Jiao, J.* (2026). Street Semantic Tree: A Knowledge-driven GeoAI Framework for Urban E-scooter Ridership Classification. *International Journal of Geographical Information Science*.
3. Jiao, J., Afroogh, S.*, **Xu, Y.**, Phillips, C. (2025). Navigating LLM Ethics: Advancements, Challenges, and Future Directions. *AI and Ethics*.

4. Wang, T., He, C., Li, H., Li, Y., **Xu, Y.**, Wang, Y., Jiao, J. (2025). HLCG: A Hierarchical Lane-Changing Gaming Decision Model for Heterogeneous Traffic Flow on Two-Lane Highways. *Transportation Research Record*.
5. Zhang, X., Zhao, X., **Xu, Y.**, Lovreglio, R.*, Nilsson, D. (2024). Situational-Aware Multi-Graph Convolutional Recurrent Network (SA-MGCRN) for Travel Demand Forecasting During Wildfires. *Transportation Research Part A: Policy and Practice*.
6. Zhang, X.*, Zhou, Z., **Xu, Y.**, Zhao, X. (2024). Analyzing Spatial Heterogeneity of Ridesourcing Demand Determinants Using Explainable Machine Learning. *Journal of Transport Geography*.
7. Jiang, S., Sun, Y., Wong, W.*, **Xu, Y.**, Zhao, X. (2024). Real-Time Urban Traffic Monitoring Using Transit Buses as Probes. *Transportation Research Record*.
8. Zhao, X.*, **Xu, Y.**, Lovreglio, R., Kuligowski, E., Nilsson, D., Cova, T. J., Wu, A., Yan, X. (2022). Estimating Wildfire Evacuation Decision and Departure Timing Using Large-Scale GPS Data. *Transportation Research Part D: Transport and Environment*.
9. Wu, A., Yan, X.*, Kuligowski, E., Lovreglio, R., Nilsson, D., Cova, T. J., **Xu, Y.**, Zhao, X. (2022). Wildfire Evacuation Decision Modeling Using GPS Data. *International Journal of Disaster Risk Reduction*.
10. Merlin, L.*, Yan, X., **Xu, Y.**, Zhao, X. (2021). A Segment-Level Model of Shared, Electric Scooter Origins and Destinations. *Transportation Research Part D: Transport and Environment*.
11. Qi, X., Ni, Y., **Xu, Y.**, Tian, Y., Wang, J., Sun, J.* (2021). Autonomous Vehicles' Car-Following Drivability Evaluation Based on Driving Behavior Spectrum Reference Model. *Transportation Research Record*.
12. Yan, X., Yang, W., Zhang, X., **Xu, Y.**, Bejleri, I., Zhao, X.* (2021). A Spatiotemporal Analysis of E-Scooters' Relationships with Transit and Station-Based Bikeshare. *Transportation Research Part D: Transport and Environment*.
13. Sun, J.*, Qi, X., **Xu, Y.**, Tian, Y. (2019). Vehicle Turning Behavior Modeling at Conflicting Areas of Mixed-Flow Intersections Based on Deep Learning. *IEEE Transactions on Intelligent Transportation Systems*.
14. Ma, Z., Xie, J., Qi, X., **Xu, Y.**, Sun, J.* (2017). Two-Dimensional Simulation of Turning Behavior in Potential Conflict Area of Mixed-Flow Intersections. *Computer-Aided Civil and Infrastructure Engineering*.

Manuscripts Under Review or Revision

1. **Xu, Y.***, Jiao, J. (2026). Exploring Autonomous Vehicle Crash Severity Using Interpretable Machine Learning and Large Language Models. *Under review (2nd round)*.
2. **Xu, Y.***, Xu, N., Zhang, X., Wang, T., Claudel, C., Jiao, J. (2026). Household-Joint Generation of Travel Patterns with a Fine-Tuned Large Language Model. *Under review*.
3. **Xu, Y.***, Jiao, J. (2026). A Knowledge Graph Guided Vision-Language Model for Safety-Critical Autonomous Driving Decisions. *Under review*.
4. Wang, T.†, **Xu, Y.**†, Byeon, J., Jiao, J., Mohammadi, J., Kockelman, K., Claudel, C.*, Bayen, A. (2026). Solar Phased Arrays-Based Wireless Power for Commercial Aviation Decarbonization. *Under review. Preprint at Research Square*.
5. Wang, T.*, Du, W., Chen, J., Zeng, T., Li, X., Byeon, J., Wang, Y., **Xu, Y.**, Wang, Y., Gao, B., Chen, S., Guo, Z., Jiao, J., Claudel, C., Bayen, A. (2026). VehDyn: A Driving World Model Benchmark for Vehicle Dynamics. *Under review. Preprint at arXiv:2609.33264*.
6. Afroogh, S.*, Pouresmaeil, Y., **Xu, Y.**, Chen, K., Murali, A., Jiao, J. (2026). From Text to Discovery: How Large Language Models Are Reshaping Research Across Scientific and Humanistic Disciplines. *Under review. Preprint at arXiv:2606.08723*.
7. Afroogh, S.*, Ahmed, S., ... **Xu, Y.** ... Zhao, J. (2026). Beyond Explainable AI (XAI): An Overdue Paradigm Shift and Post-XAI Research Directions. *Under review. Preprint at arXiv:2602.24176*.
8. **Xu, Y.***, Jiao, J., Wang, H. (2025). From Data to Decisions: An Urban Digital Twin Framework for Sustainable and Proactive Urban Management. *Under review. Preprint at SSRN 4958614*.
9. Jiao, J., Park, J., **Xu, Y.***, Sussman, K., Atkinson, L. (2025). SafeMate: A Modular RAG-Based Agent for Context-Aware Emergency Guidance. *Under review. Preprint at arXiv:2505.02306*.

Peer-Reviewed Conference Papers

1. **Xu, Y.***, Xu, N., Zhang, X., Wang, T., Claudel, C., Jiao, J. (2027). Household-Joint Generation of Travel Patterns with a Fine-Tuned Large Language Model. *Transportation Research Board 106th Annual Meeting, Washington, D.C.* (Accepted).
2. Xu, N., Zhang, X., **Xu, Y.***, Chen, X., Bredikhina, O., Jones, S. (2027). Household Fleet Contexts and Electric Vehicle Ownership and Use: A Latent-Class and Interpretable Modeling Analysis of Puget Sound Households. *Transportation Research Board 106th Annual Meeting, Washington, D.C.* (Accepted).
3. Wang, T., **Xu, Y.**, Byeon, J., Jiao, J., Mohammadi, J., Kockelman, K., Claudel, C.*, Bayen, A. (2027). Solar Phased Arrays-Based Wireless Power for Commercial Aviation Decarbonization. *Transportation Research Board 106th Annual Meeting, Washington, D.C.* (Accepted).
4. Wang, T., Chen, J., **Xu, Y.**, Li, X., Zeng, T., Chou, C., Lu, N., Peng, L., Jiao, J., Claudel, C.* (2027). RoboPhys-3D: A Comprehensive Embodied World Model Evaluation via 3D Reconstruction. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Orlando, FL.* (Accepted).
5. Wang, T., Byeon, J., Yehia, A., **Xu, Y.**, Park, J., Zeng, T., Chen, S., Wang, Z., Jiao, J., Claudel, C.* (2026). XR-DT: Extended Reality-Enhanced Digital Twin for Safe Motion Planning via Human-Aware Model Predictive Path Integral Control. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Pittsburgh, PA.*
6. Wang, T.†, **Xu, Y.**†, Byeon, J., Jiao, J., Mohammadi, J., Kockelman, K., Claudel, C.*, Bayen, A. (2026). Solar Phased Arrays-Based Wireless Power Transfer to Boost the Range of Hybrid Electric Aircraft in Solar-Rich World. *2026 INFORMS Annual Meeting, San Francisco, CA.* (Accepted).
7. **Xu, Y.***, Jiao, J. (2026). Evaluating Retrieval-Augmented Generation Strategies for Large Language Models in Travel Mode Choice Prediction. *Transportation Research Board 105th Annual Meeting, Washington, D.C.*
8. **Xu, Y.***, Jiao, J., Deng, C. (2026). Analyzing Factors Associated with Autonomous Vehicle Crash Severity Using Interpretable Machine Learning. *Transportation Research Board 105th Annual Meeting, Washington, D.C.*
9. Yehia, A., Byeon, J., Wang, T., Wang, H., **Xu, Y.**, Jiao, J., Claudel, C.* (2026). ARCAS: An Augmented Reality Collision Avoidance System with SLAM-Based Tracking for Enhancing VRU Safety. *2026 IEEE Intelligent Vehicles Symposium (IV), Detroit, MI.*
10. Lewis, S.*, Jiao, J., **Xu, Y.**, Park, J., Phillips, C. (2026). A Decentralized Digital Twin via Crowdsourced Sensing and Browser-Based Edge Computation. *AAAI 2026 Deployable AI Workshop, Singapore.* (Accepted).
11. **Xu, Y.***, Park, J., Jiao, J. (2025). LLM-Powered Digital Twins for Interactive Urban Mobility Simulation: Integrating SUMO with AI Agents. *NeurIPS 2025 UrbanAI Workshop, San Diego, CA.*
12. Jiao, J., Lewis, S.*, **Xu, Y.**, Park, J., Phillips, C. (2025). OpenCityCorpus: A Large-Scale, Harmonized, and LLM-Ready Corpus of Urban Data for Scientific Research. *NeurIPS 2025 AI for Science Workshop, San Diego, CA.*
13. **Xu, Y.***, Jiao, J. (2025). Assessing the Effects of Built Environment and Demographics on E-Scooter and E-Bike Usage on City Streets: A Case Study of Austin, TX. *Transportation Research Board 104th Annual Meeting, Washington, D.C.*
14. **Xu, Y.***, Jiao, J., Wang, H. (2025). An Urban Digital Twin Framework for Sustainable Transportation and Smart Cities: A Case Study of Austin, TX. *Transportation Research Board 104th Annual Meeting, Washington, D.C.*
15. **Xu, Y.***, Jiao, J., Li, Y. (2025). Exploring Spatial Heterogeneity of E-scooter's Relationship with Ridesourcing using Explainable Machine Learning. *Transportation Research Board 104th Annual Meeting, Washington, D.C.*
16. Xu, N., **Xu, Y.***, Liu, J., Jiao, J. (2025). How Do EV Crashes Differ from ICEV Crashes: A Comparative Study of Pennsylvania. *Transportation Research Board 104th Annual Meeting, Washington, D.C.*
17. Wang, H.*, Jiao, J., **Xu, Y.** (2025). Street Function Representation Learning on Long-Term Traffic Flow Prediction. *Transportation Research Board 104th Annual Meeting, Washington, D.C.*

18. Wang, H.*, Davis, W., Jiao, J., **Xu, Y.** (2025). Urban E-scooter Usage Prediction Based on Semantic Descriptions: A Knowledge-Driven AI. *Transportation Research Board 104th Annual Meeting, Washington, D.C.*
19. Wang, T.*, Guo, Q., He, C., Li, H., **Xu, Y.**, Wang, Y., Jiao, J. (2025). Impact of Connected and Automated Vehicles on Longitudinal and Lateral Performance of Heterogeneous Traffic Flow in Shared Autonomy on Two-Lane Highways. *WCX SAE World Congress Experience, Detroit, MI.*
20. Chio, S., **Xu, Y.***, Jiao, J. (2024). Utility or Equity? A Critical Analysis of Existing Public Electric Vehicle Charger Allocations in Austin, Texas. *Association of Collegiate Schools of Planning Annual Conference, Seattle, WA.*
21. Jiang, S., **Xu, Y.**, Wai, W., Zhao, X.* (2024). Real-Time Urban Traffic Monitoring Using Transit Buses as Probes. *Transportation Research Board 103rd Annual Meeting, Washington, D.C.*
22. **Xu, Y.**, Ke, Q., Zhao, X.* (2023). ICN: Interactive Convolutional Network for Forecasting Travel Demand of Shared Micromobility. *Transportation Research Board 102nd Annual Meeting, Washington, D.C.*
23. **Xu, Y.**, Xiong, R., Lovreglio, R., Nilsson, D., Zhao, X.* (2023). Forecasting Real-Time Travel Demand During Wildfire Evacuations: A Situational-Aware Multi-Graph Convolutional Recurrent Network (SA-MGCRN) Approach. *Transportation Research Board 102nd Annual Meeting, Washington, D.C.*
24. **Xu, Y.**, Paliwal, M., Zhao, X.* (2022). Real-Time Forecasting of Dockless Scooter-Sharing Demand: A Context-Aware Spatio-Temporal Multi-Graph Convolutional Network Approach. *Transportation Research Board 101st Annual Meeting, Washington, D.C.*
25. Zhao, X.*, **Xu, Y.**, Lovreglio, R., Kuligowski, E., Nilsson, D., Cova, T. J., Wu, A., Yan, X. (2022). Estimating Wildfire Evacuation Decision and Departure Timing Using Massive GPS Data. *Transportation Research Board 101st Annual Meeting, Washington, D.C.*
26. **Xu, Y.**, Yan, X., Sisiopiku, V., Merlin, L., Xing, F., Zhao, X.* (2021). Micromobility Trip Origin and Destination Inference Using General Bikeshare Feed Specification (GBFS) Data. *Transportation Research Board 100th Annual Meeting, Washington, D.C.*
27. Qi, X., Ni, Y., **Xu, Y.**, Tian, Y., Wang, J., Sun, J.* (2021). Autonomous Vehicles' Car-Following Drivability Evaluation Based on Driving Behavior Spectrum Reference Model. *Transportation Research Board 100th Annual Meeting, Washington, D.C.*
28. Chen, D., **Xu, Y.**, Sun, J.* (2019). Vehicle Cooperation Around Lane-Changing. *Transportation Research Board 98th Annual Meeting, Washington, D.C.*

Patents

1. Sun, J., **Xu, Y.**, Yu, R. A Road-Virtual Parallel Testing Scheme for Autonomous Vehicles. China Patent No. 201810417326.2, issued October 2018.
2. Sun, J., **Xu, Y.**, Ye, Y. A Scenario Regeneration and Accelerated Test Method for Autonomous Vehicles. China Patent No. 201710568536.7, issued October 2017.

Book Chapter, Dataset, and Technical Reports

1. **Xu, Y.** (2026). AI in Urban Infrastructure and Utilities. [Book Chapter], *Smart Cities with AI, Taylor & Francis.*
2. **Xu, Y.**, Jiao, J., Chen, Y. (2025). An Augmented Dataset of Autonomous Vehicle Collisions in California [Dataset]. *Zenodo.* https://doi.org/10.5281/zenodo.15937591.
3. Jiao, J., **Xu, Y.** (2024). Scooter-Share Travel Demand Forecast: A Context-Aware LSTM Recurrent Neural Network Approach. [Project Report], *USDOT Tier 1 UTC: Cooperative Mobility for Competitive Megaregions.*
4. Jiao, J., **Xu, Y.** (2024). Digital Twin as Catalyst for Sustainable and Smart City. [Project Report], *USDOT Tier 1 UTC: Center for Climate-Smart Transportation.*
5. Zhao, X., Sisiopiku, V., Steiner, R., **Xu, Y.**, Liu, Y., Yan, D., Khalil, J., Yang, W., Jafarzadehfadaki, M., Suarez, J. (2022). Micromobility as a Solution To Reduce Urban Traffic Congestion. [Project Report], *USDOT Regional UTC: Southeastern Transportation Research, Innovation, Development and Education Center.*

PRESENTATIONS & TALKS

1. **Xu, Y.** (2026). Generative Traveler Agents for Urban Mobility Planning. *4th Annual Good Systems Smart Cities and AI Innovations Symposium, Austin, TX.*
2. **Xu, Y.** (2026). LLM-Powered Digital Twins for Interactive Urban Mobility Simulation. *Texas Symposium on Machine Learning, Responsible AI & Robotics, Austin, TX.*
3. **Xu, Y.** (2026). Digital Twins for Smarter Cities. *Transportation Research Board Annual Meeting, Washington, D.C.*
4. **Xu, Y.** (2026). AI-Powered Transportation Digital Twin. *Transportation Research Board Annual Meeting, Washington, D.C.*
5. **Xu, Y.** (2025). Urban Digital Twins for Smart Cities. *City of Austin - Digital Twin Partnership in Action Workshop, Austin, TX.*
6. **Xu, Y.** (2025). LLM-Enabled Transportation Digital Twin for Smart Cities. *Smart Cities and AI Symposium, Austin, TX.*
7. **Xu, Y.** (2025). Digital Twins and LLM Agents for Smart Cities. *Good Systems Symposium, Austin, TX.*
8. **Xu, Y.** (2024). A Digital Twin for the City of Austin. *2nd Annual Smart Cities and AI Innovations Symposium, Austin, TX.*
9. **Xu, Y.** (2024). Where There's Fire, There's Smoke. Using AI and Digital Twins to Prepare for Climate Change. *Smart Cities Connect Conference, Austin, TX.*
10. **Xu, Y.** (2023). Real-Time Forecasting of Dockless Scooter-Sharing Demand. *UT Smart Cities Talk Series, Austin, TX.*
11. **Xu, Y.**, Paliwal, M., Zhao, X. (2021). Real-Time Forecasting of Dockless Scooter-Sharing Demand: A Spatio-Temporal Multi-Graph Convolutional Network Approach. *The 2021 TRB Workshop Sponsored by AED50, Washington, D.C. & The UF AI Research Catalyst Fund Seminar.*
12. **Xu, Y.**, Yan, X., Liu, X., Zhao, X. (2020). Applying Interpretable Machine Learning to Identify Key Factors Associated with Neighborhood Ride-Splitting Adoption Rate and to Model Their Nonlinear Relationships. *Transportation Research Board ABJ70 Committee meeting, Washington, D.C.*

RESEARCH GRANTS

Neuro-Symbolic Edge AI for AV Safety: A Knowledge-Graph Guided VLM **Mar 2026 - Mar 2027**
NVIDIA Academic Grant, Co-PI.

Digital Twins as a Catalyst for Sustainable and Smart Cities **Oct 2023 - Sep 2024**
USDOT Tier 1 UTC: Center for Climate-Smart Transportation, $241,478, Co-PI.

A Highway Vehicle Routing Dataset During the 2019 Kincade Fire Evacuation **Apr 2021 - Oct 2021**
Natural Hazards Center Weather Ready Research Award Program, $2,500, Co-PI.

TEACHING EXPERIENCE

CRP 386-14: Smart City **Fall 2026**
Co-Instructor *The University of Texas at Austin*

CRP 381Q: Quantitative Methods **Spring 2026**
Co-Instructor *The University of Texas at Austin*

UGS 302: Ethical AI: Good Systems **Spring 2025**
Guest Lecturer *The University of Texas at Austin*

BDP 319: Introduction to Smart Cities **Spring 2025**
Guest Lecturer *The University of Texas at Austin*

CRP 386: Urban Geographic Information Systems **Fall 2024**
Guest Lecturer *The University of Texas at Austin*

CRP 395D/386: Smart City Practicum **Spring 2024**
Guest Lecturer *The University of Texas at Austin*

LA 329: Global Learning Seminar **Spring 2024**
Guest Lecturer *The University of Texas at Austin*

CGN 6905: Machine Learning Applications in Civil Engineering **Spring 2021**
Teaching Assistant *University of Florida*

MENTORING EXPERIENCE

Jihyung Park **Spring 2025 - Fall 2025**
M.S. Student, Department of Computer Science *The University of Texas at Austin*

Ross Burgett **Fall 2025**
M.S. Student, School of Architecture *The University of Texas at Austin*

Zhewei Xie **Spring 2025 - Fall 2025**
M.S. Student, Department of Economics *The University of Texas at Austin*

Tianyi Wang **Summer 2024 - Summer 2025**
M.S. Student, Department of Mechanical Engineering and Materials Science *Yale University*

Claire Deng **Fall 2024 - Summer 2025**
High School Student *Westwood High School*

Yu Chen **Fall 2024 - Summer 2025**
M.S. Student, School of Architecture *The University of Texas at Austin*

Aaron Purewal **Fall 2023 - Spring 2024**
B.S. Student, McCombs School of Business *The University of Texas at Austin*

Xiaohe Yin **Fall 2023 - Spring 2024**
Research Assistant, Tandon School of Engineering *New York University*

Kay Kong **Fall 2023 - Spring 2024**
B.S. Student, Department of Computer Science *The University of Texas at Austin*

Jakob Love **Fall 2023 - Spring 2024**
B.S. Student, Department of Aerospace Engineering and Engineering Mechanics *The University of Texas at Austin*

Ruoyang Xiong **Fall 2021 - Fall 2022**
M.S. Student, Department of Computer and Information Science and Engineering *University of Florida*

Yepeng Liu **Fall 2020 - Fall 2021**
M.S. Student, Department of Computer and Information Science and Engineering *University of Florida*

Mudit Paliwal **Spring 2020 - Spring 2021**
M.S. Student, Department of Industrial and Systems Engineering *University of Florida*

Alex Wu **Spring 2020 - Fall 2021**
B.S. Student, Department of Civil and Coastal Engineering *University of Florida*

SERVICE AND PROFESSIONAL AFFILIATIONS

Journal Reviewer

- Accident Analysis and Prevention
- Applied Geography
- Case Studies on Transport Policy
- Cities
- Computers & Graphics
- Computers, Environment and Urban Systems

- Journal of Cycling and Micromobility Research
- Journal of Intelligent Transportation Systems: Technology, Planning, and Operations
- Journal of Transport Geography
- Multimodal Transportation
- PeerJ Computer Science
- Research in Transportation Economics
- Transport Policy
- Transport Reviews
- Transportmetrica A: Transport Science
- Transportation Research Interdisciplinary Perspectives
- Transportation Research Part A: Policy and Practice
- Transportation Research Part C: Emerging Technologies
- Transportation Research Part D: Transport and Environment
- Transportation Research Record
- Travel Behaviour and Society

Conference Reviewer

- Annual Conference on Innovative Applications of Artificial Intelligence (IAAI)
- COTA International Conference of Transportation Professionals
- IEEE International Conference on Intelligent Transportation Systems (ITSC)
- IEEE Intelligent Vehicles Symposium (IV)
- International Association for China Planning (IACP) Conference
- Transportation Research Board Annual Meeting (TRB)

Member

- Association of Collegiate Schools of Planning (ACSP)
- American Association of Geographers (AAG)
- Chinese Overseas Transportation Association (COTA)
- World Transportation Convention (WTC) GH0401 Technical Committee
- Institute of Transportation Engineers (ITE) Student Chapter (2019 - 2023)

HONORS & AWARDS

- Top 10 Most Cited Articles in *Transportation Research Part A: Policy and Practice*, 2021 - 2024
- Transportation Research Board Annual Meeting Travel Award, *University of Florida*, 2020, 2021, 2022, and 2023
- Graduate Academic Scholarship, *Tongji University*, 2016, 2017, and 2018
- Undergraduate Academic Excellence Scholarship, *Tongji University*, 2013